

Appendix 1: Recommended settings for different materials

OMNI 6W PARAMETER--70mm lens

Material Category	Material	Passes	Speed/Dwell Time	Freq (KHz)	Pulse-Width	Interval	Note	
	3D crystal		800	40	12			
Glass/Ceramics	Glass Engraving	1	400	40	1	0.02		
	Glass Deep Engraving/Matte Glass	1	3200	40	2	0.01		
	Glass Cutting(0.3mm)	50	200	40	1			
	Ceramic	2	1000	40	1	0.02		
Metal	Aluminum Engraving	1	2000	40	1	0.02		
	Aluminum Photos	1	100 us	40	12	DPI:850	image mode: Halftone	negative image
	Aluminum Cutting (0.5mm)	12	10	40	1			
	Brass Engraving	1	1000	40	1	0.02		
	Brass (Deep Brown)	1	200	30	6	0.01		
	Brass Photos	1	100 us	40	1	DPI:850	image mode: Halftone	
	Stainless Steel Engraving	1	2000	40	1	0.02		
	Stainless Steel Photos	1	100 us	40	6	DPI:850	image mode: Halftone	negative image
	Stainless Steel(Brown)	1	600	30	1	0.02		
	Stainless Steel Photo(Brown)	1	500 us	40	1	DPI:2550	image mode: Halftone	
	Stainless Steel(Black)	1	20	40	1	0.01		Defocus 10mm
	Mirror Stainless Steel Engraving	1	2000	40	1	0.02		
	Mirror Stainless Steel Photos	1	400 us	40	12	DPI:850	image mode: Halftone	negative image
	Painted Metal Engraving	1	2000	40	1	0.02		
	Painted Metal Photos	1	400 us	40	14	DPI:850	image mode: Halftone	
Tumblers	White Coated Tumblers (Remove Coat)	3	1500	40	1	0.01		Defocus 10mm
	Black Coated Tumblers (Remove Coat)	2	1600	50	1	0.01		Defocus 10mm
	Black Coated Tumblers (Brown/Gold)	3	700	30	1	0.02		Defocus 10mm
Acrylic	Black Acrylic Engraving	1	2000	40	1	0.02		
	Black Acrylic Photos	1	400 us	40	14	DPI:850		negative image
	Black Acrylic Cutting(3mm)	6	10	40	1			
	Transparent Acrylic Engraving	1	1000	40	1	0.02		
	Transparent Acrylic Photos	1	400 us	40	1	DPI:850	image mode: Halftone	
	Transparent Acrylic Lamp Base	2	500	30	1	0.02		
	Colorful Acrylic Engraving	1	1600	30	4	0.02		
	Colorful Acrylic Photos	1	500 us	40	6	DPI:850	image mode: Halftone	
	Colorful Acrylic Cutting(3mm)	55	300	30	1			
Plastic/Silicone	PC/ABS	1	2000	40	18	0.02		
	PC Plastic Cutting (0.5mm)	5	200	40	1			
	Nylon	1	4000	40	12	0.05		
	TPU	1	1200	40	1	0.01		
	PET/PI (PET Film)	1	4400	30	1	0.02		
	PVC	1	2000	30	12	0.06		
	Self-adhesive labels	1	2000	30	12	0.06		
	Silicone	1	1200	40	1	0.01		
	Rubber	1	3000	30	6	0.02		

OMNI 6W PARAMETER--70mm lens

Material Category	Material	Passes	Speed/Dwell Time	Freq (KHz)	Pulse-Width	Interval	Note	
Wood	Wood Engraving	2	2000	40	6	0.02		
	Wood Photos	1	100 us	40	1	DPI:1250	image mode: Halftone	negative image
	Wood Dark Color	1	200	40	6	0.02		
	Cutting Plywood(3mm)	15	80	40	1			
	Wood (3D Engraving)	depends on the depth	1000	40	1	0.02		
	Wood--Walnut	1	400	40	1	0.01		
	Bamboo	1	400	30	4	0.01		
	Cork Coasters Engraving	1	1600	40	6	0.02		
	Cork Coasters (Deep Engraving)	1	600	40	1	0.02		
	Cork Coasters Photos	1	300 us	40	6	DPI:2550	image mode: Halftone	
Fabric	Leather Marking	1	2000	30	24	0.02		
	Leather(Black Marking)	2	1500	40	1	0.02		
	Black Leather(White)	1	6000	35	18	0.02		
	Silk	1	2000	40	18	0.02		
	Jean	1	2000	40	8	0.02		
	Towels	1	5000	30	20	0.02		
	Cutting Felt	3	400	40	1			
	Black cotton T-shirt	1	2000	40	8	0.02		
Electronic Products	Phone Cases (TPU)	1	1200	40	1	0.01		
	Charger Adaptor	1	2000	40	18	0.02		
	PCB Marking	1	2000	40	18	0.02		
	USB	1	2000	40	18	0.02		
Food	Orange	2	2000	40	1	0.02		
	Apple	1	2400	40	1	0.01		
	Banana	1	1400	40	1	0.02		
	Macarons	3	800	30	1	0.03		
	Egg	1	2000	40	1	0.02		
	Chocolate	1	3000	30	6	0.02		
	Biscuit	1	2400	30	6	0.02		
Sport Products	Baseball	1	2000	30	18	0.02		
	Table Tennis Ball	1	3000	30	10	0.02		
	Golf Ball	1	1400	40	24	0.02		
Stone/Slate	Slate Marking(Grey)	1	2000	40	1	0.02		
	Slate Marking (White)	1	200	40	1	0.03	defocus 15mm	
	Slate 3D Engraving	120	1400	40	1	0.02		
	Stone Marking	9	1000	40	1	0.02		
	Stone deep Engraving	256	1000	40	1	0.02	4mm depth	
Paper	Paper Marking	1	2000	30	6	0.02		
	Paper Cutting	2	200	40	1			
	Paper Photos	3	300 us	40	1	DPI:1250	image mode: Halftone	
Others	Leaf Photos	1	100 us	40	1	DPI:1250	image mode: Halftone	negative image
	Black Carbon Fiber Marking	1	3000	40	18	0.02	defocus 10mm	
	Carbon Fiber Cutting (0.2mm)	4	200	40	1			

OMNI 6W PARAMETER--150mm lens

Material Category	Material	Passes	Speed/Dwell Time	Freq (KHz)	Pulse-Width	Interval	Note	
Glass/Ceramics	Glass Engraving	1	200	40	1	0.03		
	Glass Deep Engraving/Matte Glass	1	1600	40	2	0.02		
	Glass Cutting(0.3mm)	50	100	40	1			
	Ceramic	2	500	40	1	0.03		
Metal	Aluminum Engraving	1	1000	40	1	0.03		
	Aluminum Photos	1	400 us	40	10	DPI:400	image mode: Halftone	nagetive image
	Aluminum Cutting (0.5mm)	12	5	40	1			
	Brass Engraving	1	500	40	1	0.03		
	Brass (Deep Brown)	1	100	30	5	0.015		
	Brass Photos	1	400 us	40	1	DPI:400	image mode: Halftone	
	Stainless Steel Engraving	1	1000	40	1	0.03		
	Stainless Steel Photos	1	400 us	40	5	DPI:400	image mode: Halftone	nagetive image
	Stainless Steel(Brown)	1	300	30	1	0.03		
	Stainless Steel Photo(Brown)	1	500 us	40	1	DPI:1000	image mode: Halftone	
	Stainless Steel(Black)	1	10	40	1	0.01		Defocus 10mm
	Mirror Stainless Steel Engraving	1	1000	40	1	0.03		
	Mirror Stainless Steel Photos	1	500 us	40	10	DPI:400	image mode: Halftone	nagetive image
	Painted Metal Engraving	1	1000	40	1	0.03		
	Painted Metal Photos	1	500 us	40	12	DPI:400	image mode: Halftone	
Tumblers	White Coated Tumblers (Remove Coat)	3	750	40	1	0.015		Defocus 10mm
	Black Coated Tumblers (Remove Coat)	2	800	50	1	0.01		Defocus 10mm
	Black Coated Tumblers (Brown/Gold)	3	350	30	1	0.03		Defocus 10mm
Acrylic	Black Acrylic Engraving	1	1000	40	1	0.03		
	Black Acrylic Photos	1	500 us	40	12	DPI:400		nagetive image
	Black Acrylic Cutting(3mm)	6	5	40	1			
	Transparent Acrylic Engraving	1	500	40	1	0.03		
	Transparent Acrylic Photos	1	400 us	40	1	DPI:400	image mode: Halftone	
	Transparent Acrylic Lamp Base	2	250	30	1	0.03		
	Colorful Acrylic Engraving	1	800	30	3	0.03		
	Colorful Acrylic Photos	1	500 us	40	5	DPI:500	image mode: Halftone	
	Colorful Acrylic Cutting(3mm)	55	150	30	1			
Plastic/Silicone	PC/ABS	1	1000	40	15	0.03		
	PC Plstic Cutting (0.5mm)	5	100	40	1			
	Nylon	1	2000	40	10	0.1		
	TPU	1	600	40	1	0.025		
	PET/PI (PET Film)	1	2200	30	1	0.03		
	PVC	1	1000	30	10	0.1		
	Self-adhesive labels	1	1000	30	10	0.1		
	Silicone	1	600	40	1	0.025		
	Rubber	1	1500	30	5	0.03		

OMNI 6W PARAMETER--150mm lens

Material Category	Material	Passes	Speed/Dwell Time	Freq (KHz)	Pulse-Width	Interval	Note	
Wood	Wood Engraving	2	1000	40	5	0.03		
	Wood Photos	1	200 us	40	1	DPI:850	image mode: Halftone	nagetive image
	Wood Dark Color	1	100	40	5	0.03		
	Cutting Plywood(3mm)	15	40	40	1			
	Wood (3D Engraving)	depends on the depth	500	40	1	0.03		
	Wood--Walnut	1	200	40	1	0.02		
	Bamboo	1	200	30	3	0.02		
	Cork Coasters Engraving	1	800	40	5	0.03		
	Cork Coasters (Deep Engraving)	1	300	40	1	0.03		
	Cork Coasters Photos	1	400 us	40	5	DPI:1250	image mode: Halftone	
Fabric	Leather Marking	1	1000	30	20	0.03		
	Leather(Black Marking)	2	750	40	1	0.03		
	Black Leather(White)	1	3000	35	15	0.03		
	Silk	1	1000	40	15	0.03		
	Jean	1	1000	40	7	0.03		
	Towels	1	2500	30	17	0.03		
	Cutting Felt	3	200	40	1			
	Black cotton T-shirt	1	1000	40	7	0.03		
Electronic Products	Phone Cases (TPU)	1	600	40	1	0.025		
	Charger Adaptor	1	1000	40	15	0.03		
	PCB Marking	1	1000	40	15	0.03		
	USB	1	1000	40	15	0.03		
Food	Orange	2	1000	40	1	0.03		
	Apple	1	1200	40	1	0.02		
	Banana	1	700	40	1	0.03		
	Macarons	3	400	30	1	0.05		
	Egg	1	1000	40	1	0.03		
	Chocolate	1	1500	30	5	0.03		
	Biscuit	1	1200	30	5	0.03		
Sport Products	Baseball	1	1000	30	15	0.03		
	Table Tennis Ball	1	1500	30	8	0.03		
	Golf Ball	1	700	40	20	0.03		
Stone/Slate	Slate Marking(Grey)	1	1000	40	1	0.03		
	Slate Marking (White)	1	100	40	1	0.05	defocus 15mm	
	Slate 3D Engraving	120	700	40	1	0.03		
	Stone Marking	9	500	40	1	0.03		
	Stone deep Engraving	256	500	40	1	0.03	4mm depth	
Paper	Paper Marking	1	1000	30	5	0.03		
	Paper Cutting	2	100	40	1			
	Paper Photos	3	400 us	40	1	DPI:650	image mode: Halftone	
Others	Leaf Photos	1	200 us	40	1	DPI:850	image mode: Halftone	nagetive image
	Black Carbon Fiber Marking	1	1500	40	15	0.03	defocus 10mm	
	Carbon Fiber Cutting (0.2mm)	4	100	40	1			